Smart City Traffic Flow Simulator

README

About

A Python-based traffic simulation project that optimizes traffic signal timing, prioritizes ambulances, predicts traffic congestion, and generates alerts to support smarter urban traffic management.

Features

- Adaptive traffic signal timing
- Ambulance priority simulation
- Traffic congestion prediction
- Traffic alerts and classification
- Data visualization using graphs

Technologies

Python • NumPy • Pandas • Matplotlib • Jupyter Notebook

Installation

Install the required libraries:

```
pip install numpy pandas matplotlib jupyter
```

How to Run

1. Clone or download this repository.
2. Open the project notebook in Jupyter Notebook.
3. Run all cells to see the simulation results.

Objective

To demonstrate how data analysis and intelligent signal management can help reduce traffic congestion and improve emergency vehicle movement.

Future Improvements

- Real-time traffic monitoring
- Machine learning-based prediction
- Smart city traffic dashboard
- Ambulance route optimization

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